

Chapter 12 — Emerging trends

The chapter closes by looking ahead

Learning objectives

- By the end of this part

Directions still shifting operations

- Section 12.2 already covered cloud object stores, SQL versus NoSQL trade-offs
- Automated pipelines that keep storage and analytics synchronized

Example 12.26 — Serverless Compute Over Object Storage

- Example 12.26 — hands-on module
- Beyond managed buckets themselves
- Example 12.26 pairs S3-class storage with serverless compute
- Explore the chapter example module
- View files: ``modules/chapter12/example26/``

Example 12.27 — Neo4j for Relationship-Heavy Graphs

- Example 12.27 — hands-on module
- Horizontal NoSQL stores remain important
- Example 12.27 places graph databases in network-style apps
- Graph databases such as Neo4j fit social graphs and fraud rings where multi-hop
- Explore the chapter example module
- View files: ``modules/chapter12/example27/``

Example 12.28 — InfluxDB for Timestamped Metrics

- Example 12.28 — hands-on module
- Example 12.28 places time-series stores in monitoring workloads
- Time-series databases such as InfluxDB store high-rate sensor and market ticks for
- Purpose-built engines avoid forcing every metric into a general document or relational
- Explore the chapter example module
- View files: ``modules/chapter12/example28/``

Example 12.29 — Kafka High-Throughput Messaging

- Example 12.29 — hands-on module
- Chapter 9 covers streaming collection
- Example 12.29 returns to Kafka in the pipeline-automation context
- Explore the chapter example module
- View files: ``modules/chapter12/example29/``

Looking ahead to Chapter 13

- Scalable management provides the storage, consistency, versioning
- Chapter 13 turns to reproducibility and open science
- Emerging tools matter only when they still expose addressable versions and auditable

Takeaways

- Serverless compute scales independently of durable object storage
- Neo4j and InfluxDB illustrate purpose-built graph and time-series engines
- Kafka remains a high-throughput messaging backbone for live pipelines
- Trends build on earlier foundations rather than replacing CAP, formats, versioning
- Chapter 13 consumes the managed assets this chapter prepares

Next

- Complete the quiz for this part
- Complete the quiz for this part to finish Chapter 12